

Process Scheduling Algorithms

Subject: Operating Systems Topic: Process Management

Overview

FCFS, SJF, Round Robin, Priority scheduling with Gantt charts

Scheduling goals

- A scheduler chooses a ready process to use the CPU; goals include throughput, response time, waiting time, and fairness.
- Turnaround time is completion time minus arrival time; waiting time is turnaround minus CPU burst for a single-burst example.
- Preemptive algorithms can interrupt a running process; non-preemptive algorithms wait until it blocks or finishes.

Common algorithms

- FCFS is simple and non-preemptive but can cause the convoy effect when a long job blocks short jobs.
- Shortest Job First minimizes average waiting time when burst lengths are known; its preemptive form is shortest remaining time first.
- Round Robin gives each ready process a time quantum and is responsive for interactive workloads.
- Priority scheduling selects by priority; aging can reduce starvation by gradually increasing waiting processes' priorities.

Gantt chart example

- For bursts $P1=5$, $P2=3$, $P3=1$ arriving together, FCFS order $P1, P2, P3$ gives waiting times 0,5,8.
- Round Robin with quantum 2 rotates unfinished processes after each two units, improving initial response at the cost of context switches.
- Always state arrival times, tie-breaking, preemption rules, and the quantum when calculating results.

Evaluation

- Compute per-process metrics from the schedule, then report averages without hiding starvation or response behavior.
- A very small quantum increases context-switch overhead; a very large quantum approaches FCFS.
- Real operating systems combine policies and account for I/O blocking and multiple cores.